Fabian Espinoza de Osambela

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Education

Louisiana State University 2025–present
Ph.D. in Mathematics
Working with Jerome Hoffman, Fang-Ting Tu, and Scott Baldridge.
Passed qualifying exams in algebra, real analysis, and topology, 2026.
Master's thesis in progress on syntomic cohomology.

Johns Hopkins University 2025
B.A. in Mathematics and Physics (double major)
General honors; departmental honors in physics.

Research Interests

Arithmetic geometry in mixed characteristic: curves and their Jacobians, class groups, Iwasawa theory; p-adic Hodge theory: syntomic and prismatic cohomology; homotopy theory: spectral algebraic geometry, topological Hochschild and cyclic homology; automorphic forms.

Honors and Awards

- Huel D. Perkins Doctoral Fellowship, Louisiana State University 2025–present
- Undergraduate teaching award, JHU Mathematics 2025
- General honors and departmental honors in physics, Johns Hopkins University 2025
- Dean's List, Johns Hopkins University Spring 2024, Fall 2024

Papers

1. F. Espinoza, M. S. Im, and M. Khovanov, *Two-dimensional topological quantum field theories of rank two over Dedekind domains*, Contemp. Math. **834** (2026), 33–52. arXiv:2502.04502.

In preparation

2. F. Espinoza de Osambela, *Cyclic Frobenius algebra sheaves on Dedekind schemes: rigidity, spin structures, Galois descent, and the torsion coset principle.* Expected November 2026.
3. F. Espinoza de Osambela, *Iwasawa stability of the arithmetic torsion coset.* Expected December 2026.
4. F. Espinoza de Osambela, J. Hoffman, and F.-T. Tu, *Genus-2 curves with elliptic subcovers.*

Expository Notes

- *Automorphic forms on the Heisenberg group.* Course project, revised 2026.
- *Spheres and homotopy theory.* Course project, revised 2026.
- *The Grothendieck spectral sequence.* Course project, revised 2026.

Available at fabes124.github.io/publications.

Research Experience

Frobenius algebra sheaves and genus-2 curves 2025–present
Louisiana State University. With Jerome Hoffman, Fang-Ting Tu, and Scott Baldridge
Classification of self-dual Frobenius algebra sheaves on curves and rings of integers and their behavior in $\mathbb{Z}_2$-extensions; explicit genus-2 curves with split Jacobians.

Topological quantum field theories over algebraic field extensions 2024–2025
Johns Hopkins University. Mentors: Mikhail Khovanov and Mee Seong Im
Frobenius algebra structures over Dedekind domains via projective modules; led to the published paper above.

Earlier undergraduate research, Johns Hopkins University

- Drinfeld centers of fusion 2-categories and their Morita classes; mentor Thomas Waddleton (Ibou Bah's group) 2024
- Von Neumann algebras in mathematical physics, with de Sitter space as a case study; mentor Ibou Bah 2023
- X-ray emission from Galactic black holes and its detectability by eROSITA; mentor Marc Kamionkowski 2023

Graduate Coursework

Louisiana State University. Algebra I; Real Analysis I; Topology I; Algebraic Topology; Operator Theory, Category Theory; Rational Homotopy Theory; Algebraic Geometry; Harmonic Analysis in Phase Space; Quantum Field Theory.

Johns Hopkins University. Algebra I–II; Algebraic Topology I–II; Complex Variables; Riemann Surfaces; Number Theory I; Algebraic Geometry I–II; Lie Groups and Lie Algebras; Riemannian Geometry II. Physics: Quantum Field Theory I–II; General Relativity; Advanced Particle Theory.

Reading Courses

Louisiana State University

- Deformation theory and formal moduli problems (Lurie, Pridham); class field theory; supersymmetry and Morse theory Fall 2026
- Scheme theory and sheaf cohomology; algebraic number theory (number fields); differential geometry and mathematical physics; fusion 2-categories 2025–2026

Johns Hopkins University

- Algebraic geometry: varieties, schemes, and algebraic surfaces, with Vyacheslav Shokurov Spring 2024-Spring 2025
- Characteristic classes (Milnor–Stasheff), with Akira Tominaga. Directed Reading Program Spring 2023
- Differential geometry and topology (Nakahara), with Milton Lin. Directed Reading Program Fall 2022

Talks and Posters

- *Renormalization group and Ricci flow* (poster). Undergraduate Poster Session, Department of Physics and Astronomy, Johns Hopkins University, April 2025.

- *Topological quantum field theories over rings of integers of number fields.* Knots in Washington, George Washington University, December 2024.
- *Topological quantum field theories over algebraic field extensions.* Fall 2024 Regional Undergraduate Math Research Conference, Towson University, November 2024.
- *Drinfeld center of 2-fusion categories.* Spring 2024 Regional Undergraduate Math Research Conference, Towson University, April 2024.
- *Fractional quantum Hall effect* (poster). Undergraduate Poster Session, Department of Physics and Astronomy, Johns Hopkins University, April 2024.
- *Introduction to gauge theory using classical field theory.* Society of Physics Students, Johns Hopkins University, November 2023.

Teaching

Graduate Reading Course Leader, Louisiana State University 2025–present
Led four reading courses for graduate students under the supervision of two faculty members: three in differential geometry and mathematical physics, one in algebraic geometry. Chose the material and directed the meetings.

Teaching Assistant, Mathematics, Johns Hopkins University 2023–2025
Calculus III (Fall 2023); Methods of Complex Analysis (Spring 2024, Fall 2024); Differential Equations and Applications (Spring 2025). Led weekly problem-solving sections, wrote quizzes, ran review sessions, and graded. Received the department's undergraduate teaching award.

Learning Assistant, Physics and Astronomy, Johns Hopkins University 2021–2025
Active-learning sections of Physics I (Fall 2021, Fall 2022) and Physics II (Spring 2022, Fall 2023) for physical sciences majors; Electromagnetic Theory I (Spring 2025).

Tutor, GoPeer 2021–2022
AP and college calculus and physics.

Service, Mentoring, and Leadership

- Founder and Director, Physics Undergraduate Reading Program, JHU Society of Physics Students 2023–2025
  Modeled on the math Directed Reading Program, for first- and second-year students preparing for research. Led reading courses in tensor calculus, differential topology, category theory, symmetry in particle physics, string theory, and continuum mechanics.
- Mentor and mentee, JHU physics mentorship program; organized review sessions for Advanced Classical Mechanics and Quantum Mechanics 2023-2025
- Academic mentor to about thirty undergraduates in mathematics and physics (course planning, research interests, finding advisors) 2023–2025
- Vice-President, JHU Math Club 2023–2025
- Treasurer, JHU Society of Physics Students (named a Distinguished Chapter) 2022–2023

Skills

Python, Mathematica, MATLAB, Sage, Magma, PARI.